April 2024

Econ 2042: Statistics for Economists

Tutorial Exercise III:

1). Suppose two dice are rolled. Let X be the absolute value of the difference between the two numbers showing.

a). Find the probability mass function for X.

b). Calculate the probability that X lies between 2 and 4.

c). Calculate the probability that X is less than 3 or between 2 and 4.

2). Roll two dice. Let X be equal to the number showing on the first die as proportion of the total showing. Find the probability mass function for X.

3). Suppose two coins are tossed. Let X be the number of heads that show.

a). Find the probability mass function for X.

Do the same for the case in which 4 coins are tossed.

b). Find also the distribution function for X in this case.

4). Three coins are tossed five times. Find the discrete density function for X where X equals the number of times no head appears.

5). For each of the above problems, find $E(X)$ and $var(X)$.

6). Let $f(x) = \begin{cases} c\,(2/3)^x, & x = 1, 2, 3, \ldots \\ 0, & \text{elsewhere} \end{cases}$. Find the constant c such that f is a discrete density function.

7). For each of the following probability density functions for a random variable X compute $\Pr(|X| < 1)$ and $\Pr(X^2 < 9)$.

a). $f(x) = \begin{cases} x^2/18, & -3 < x < 3 \\ 0\ , & \text{elsewhere} \end{cases}$

b). $f(x) = \begin{cases} (x+2)/18, & -2 < x < 4 \\ 0 \quad , & \text{elsewhere} \end{cases}$

8). Let $f(x) = \begin{cases} 1/2, & -1 < x < 1 \\ 0 \quad , & \text{elsewhere} \end{cases}$, be the density function of a random variable X. Define the random variable $Y = X^2$. Find the probability density function for Y. (Hint: first find the distribution function for Y).

9). A bomber drops its load on a railway track. If a bomb falls within 40 feet of the track, traffic is disrupted. Let X denotes the distance from the track that a bomb falls. Assume $f(x) = \begin{cases} (100-x)/5000, & 0 < x < 100 \\ 0 \quad , & \text{elsewhere} \end{cases}$

a). Find the probability that one bomb will disrupt traffic.

b). Find the Probability that three bombs are required to disrupt traffic.

10). X is a random variable with $E(X) = \mu$, $var(X) = \sigma^2$. Prove that if $Y = (X-\mu)/\sigma$, then $E(Y) = 0$ and $var(Y) = 1$.

11). Prove that if a and b are constant, $var(a+bx) = b^2 var(X)$.